Photometric Analysis of NGC 6946: Stellar Mass estimation

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ABSTRACT

In this report, I present a photometric analysis of the galaxy NGC 6946, based on data collected with the ToBi telescope in the r, i, and g bands. The original data were reduced and calibrated in order to obtain the most accurate estimates of the flux emitted by the galaxy in the different bands. The aim of the project is to estimate the total stellar mass of the galaxy using the empirical mass–luminosity relation proposed by Bell et al. (2003). In particular, I compared stellar mass estimates derived from mass-to-light ratios computed using two different color indices: $(g - r)$ and $(g - i)$. The results show that the total stellar mass estimated using the $(g - i)$ index is larger than the one obtained from $((g - r)$ by a factor of approximately 1.16. This difference is mainly attributed to the number of stars taken into account and to the lower sensitivity to dust extinction of the i band, which makes it a more reliable tracer of the total stellar mass in galaxies with high star formation rates, such as NGC 6946.

1 INTRODUCTION

Estimating the stellar mass of galaxies is a fundamental step in understanding their structure, dynamics and evolutionary history, as mass determines the gravitational potential and regulates key physical processes such as star formation and chemical enrichment. However, stellar mass is not a directly observable quantity: telescopes measure only the luminous flux and converting luminosity into mass requires assumptions about the mass-to-light ratio (M/L).

This quantity strongly depends on the stellar population: young massive stars are highly luminous but with a overall short life, contributing significantly to the observed light but little to the total mass. In contrast, older low-mass stars are less luminous but constitute most of the galaxy's mass. Consequently, we focus on the color of the galaxy, which provides a proxy for the age of its stellar population: blue indicate ongoing star formation and low M/L values, while red is associated with an older population and a higher M/L.

In this work, we analyze multi-band data obtained with SDSS-like filters (g, r and i). The goals of this study are two: first, to implement a complete data analysis, converting raw FITS files into calibrated science ones in CGS units, and second, to estimate the total stellar mass of the target galaxy using empirical relations between color and the mass-to-light ratio.

The target of this study is NGC 6946, a nearby ($\approx$ 5.5 Mpc) and nearly face-on spiral galaxy with a high star formation rate. These characteristics make it an ideal canditate for testing different stellar mass estimators. In particular, we compare mass estimates derived from the color $(g - r)$ with those obtained from the color $(g - i)$. The goal of this comparison is to identify the trade-off between these two approaches, in order to understand which estimator provides a more robust measurement of the total stellar mass.

2 METHODS AND DATA REDUCTION

2.1 Instrumental Noise

The raw data analyzed in this work were acquired using the optical telescope at Milano Bicocca (ToBi). The observations were carried using a set of broad-band filters (figure 1) approximating the SDSS photometric system: g (green), r (red), and i (infrared). For each filter, we had 3 to 4 available frames.

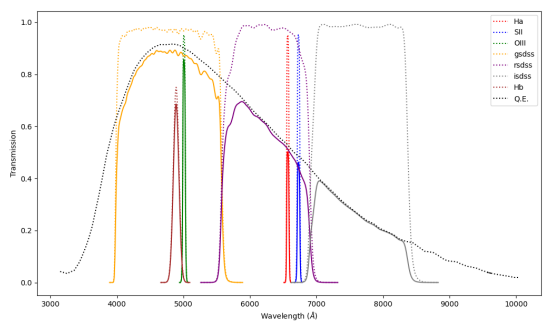

Figure 1. Transmission curves of the filter set used for the observations.

These filters were chosen to sample different stellar populations within the galaxy: as already mentioned in the introduction, the g band is sensitive to younger and hotter stars, while the r and i bands trace the older stellar population.

Since ground based observations are subject to instrumental noise and atmospheric effects, the raw images require a calibration process before any analysis can be performed. In order to subtract the electronic offset and the thermal noise, we generated respectively a Master Bias and a Master Dark frame (by averaging individual frames). Additionally, we created a normalized Master Flat frame to correct the difference in the pixel response across a flat field of view. These calibrations were applied to the science frames using the following equation:

$$\text{Science } [\mathrm{e}^-/\mathrm{s}] = \frac{\text{Raw} - (\text{Master BIAS} + \text{Master DARK})}{\text{Master Flat}} \cdot \frac{\text{Gain}}{\mathrm{t}_{\text{exposure}}} \tag{1}$$

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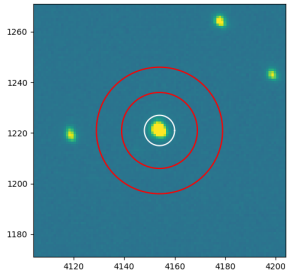

Figure 2. Aperture photometry applied to the GAIA star for a frame taken with thr *r* filter.

2.2 Photometric Calibration

Photometric calibration is essential to express the data in physical units. For each frame, we computed different calibration constants that convert the frame's flux from e^-/s to CGS units ($\mathrm{erg} \cdot \mathrm{s}^{-1} \cdot \mathrm{cm}^{-2} \cdot \mathrm{Å}^{-1}$). The method to compute these constants is simple: we make use of a star contained in the field of view of the data, with a known spectrum in the relevant filters, available from the GAIA catalog (van Leeuwen et al. (2022)), from which one derives the flux in physical units. Then we compute its experimental flux (in e^-/s) locating the star in the calibrated frames and performing aperture photometry with local sky background subtraction in a small annulus containing it (figure 2).
The calibration constant can straightforwardly be obtained by dividing the known flux in physical units by the experimental flux in e^-/s and our frames in CGS units can easily be obtained using the inverse relation applied to the full field of view of the data:

$$\mathrm{data[CGS]} = \mathrm{data}[e^-/s] \cdot \mathrm{calibration\ constant} \tag{2}$$

2.3 Sky Subtraction

We still need to address the atmospheric effects that influence our data. In fact, our processed images contain the contribution of the sky in every pixel, resulting in a luminosity that is higher than the real one. Therefore, it is necessary to subtract this sky contribution.
The first adjustment concerns the edges of the image: if they are too noisy (this can happen because the filters are smaller than the field of view), the first thing to do is to cut the image (this was necessary for the *i* filter). Furthermore, we perform a first estimate of the sky level by subtracting the median value from the image, which shifts the entire image level to be centered at zero.
However, the background is rarely perfectly uniform: this is where the detailed sky subtraction takes place. We make use of `sep`, the Python implementation of the SExtractor algorithm. This tool identifies and masks the sources above a certain threshold to compute a background model, adjusted to the non-uniformity of the sky. Before applying this tool, it is crucial to manually mask the galaxy, which is the largest source of luminosity, to prevent the algorithm from overestimating the sky level in that region. Once we apply the full sky subtraction process, we are left with an clear image of the galaxy, with a background noise reduced to the lowest possible value (figure 3).

2.4 Combining the frames

At this stage, for each filter, there are multiple science frames that are calibrated and sky subtracted. In order to reduce the noise and therefore improve the S/N ratio, we want to combine these images by averaging them. Since the individual images are not aligned, we first reproject them into a common reference frame and then we perform the combination. This provides a master FITS file for every filter that contains the cleanest and most reliable estimate of the luminous flux of the field of view (figure 4).

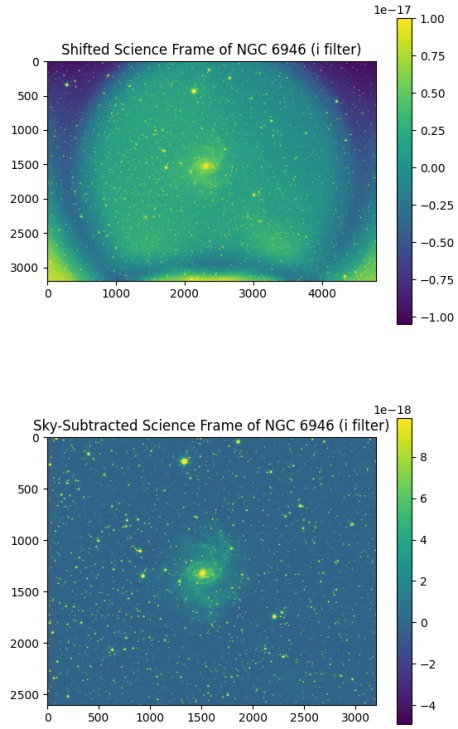

Figure 3. On the top, the shifted calibrated image of a frame in the i filter (still needs to be sky subtracted). On the bottom, the final cut and sky subtracted image.

3 MASS ESTIMATION

3.1 Flux measurement and error

The total integrated flux of NGC 6946 in the three broad line bands can be obtained by performing, once again, aperture photometry. From the final co-added images, defining an aperture that contains the totality of the galaxy, the flux can be obtained through the sum of the pixel values within this aperture.
The associated error depends on the contributes of the background noise σ_{sky} and of the source noice σ_{f}, such that

$$\sigma_{\mathrm{tot}} = \sqrt{\sigma_{\mathrm{sky}}^2 + \sigma_{\mathrm{f}}^2}\ . \tag{3}$$

The background error was estimated through a *sigma clipping* algorithm, in order to reject abnormal pixels that deviate significantly from the mean value. The source error, instead, follows the Poisson statistics and is thus simply given by the square root of the total signal collected (in electrons), later converted in CGS units, while taking account of the time of exposure and the number of frames that were used to obtain the final co-added image.

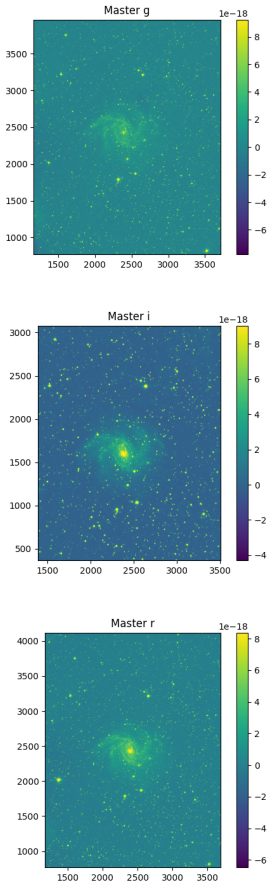

Figure 4. The three final images obtained for each filter after data reduction, sky subraction and frame co-adding

Filter	Flux [erg s^{-1} cm^{-2} A^{-1}]	Error
g	$1.4003 \cdot 10^{-12}$	$1.5757 \cdot 10^{-15}$
r	$1.8237 \cdot 10^{-12}$	$1.1193 \cdot 10^{-15}$
i	$1.8663 \cdot 10^{-12}$	$9.6981 \cdot 10^{-16}$

Table 1. Flux of the galaxy in g, r and i bands.

Table 1 presents the resulting fluxes and their relative errors in the g, r and i bands.

3.2 Mass estimate calculation

As outlined in the introduction, we computed the mass of the galaxy estimates using two different color indices, $(g-i)$ and $(g-r)$, to test the consistency of our results across different band ranges. To convert the luminous flux into stellar mass, we adopted the mass-to-light ratio relations derived by Bell et al. (2003). These relations give the following correlation between the M/L ratio in a specific band and the galaxy's color:

$$\log_{10}(M/L_{\rm b}) = a_{\rm b} + b_{\rm b}(\text{color}) \tag{4}$$

where the subscript b indicates the band we are considering, which means that obviously L, a and b will depend on the used filter. To apply this relation, we first converted the measured fluxes into luminosities. This step requires an assumption about the galaxy's distance:

Color	Band	Mass Estimate ($10^{10} M_\odot$)	Mean Mass ($10^{10} M_\odot$)
$(g-r)$	g	5.97 ± 0.03	**5.88**
	r	5.79 ± 0.02	
$(g-i)$	g	6.84 ± 0.02	**6.81**
	i	6.78 ± 0.01	

Table 2. Mass estimates. The ratio between the mass in the $(g-i)$ color and the $(g-r)$ color is 1.16

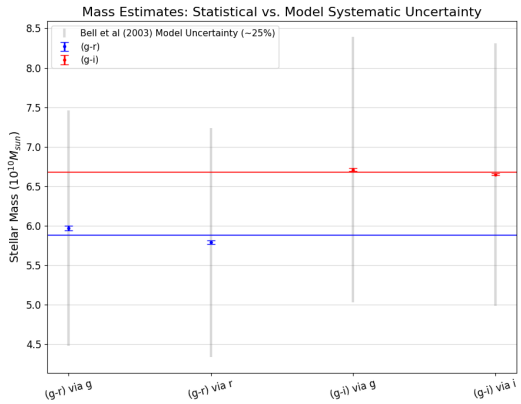

Figure 5. Comparison between the mass estimates: while the statistical error is too slow to make the results overlap, the systematic error from Bell et al. (2003) makes them compatible

we adopted a distance value of $d \approx 5.5$ Mpc, obtained by averaging the values of NGC 6946 distance given by the NED Database. Furthermore, the conversion from physical flux (in CGS units) to AB magnitudes requires the knowledge of the central wavelengths of the filters used. For the ToBi telescope filter set (figure 1), these are given by:

- $\lambda_g = 4793.0$ Å
- $\lambda_r = 6183.0$ Å
- $\lambda_i = 7655.6$ Å

The error on the mass was computed through standard error propagation: using the flux's error as the starting point, we first propagated the error of the magnitude (in logarithmic scale), then on the color indices (adding in quadrature) and finally on the luminosity and M/L ratio. The final uncertainty on the mass is given by

$$\sigma_{\rm M} = \sqrt{\left(\frac{\sigma_{\rm L}}{L}\right)^2 + \left(\frac{\sigma_{\rm M/L}}{M/L}\right)^2} \cdot M \tag{5}$$

3.3 Mass estimates results

Table 2 summarizes the results of the mass calculation. All the four obtained values reveal a mass of the order of 10^{10} solar masses. However, clear differences arise when comparing the results in detail. The discrepancy between the estimated mass in the two colors will be discussed in the next session. If we just focus on the results obtained through different bands for the same color, the value might seem similar at first glance, but their compatibility is clearly weak. This is shown in figure 5 as well, where the error bars are so little that the results do not overlap. Why does this discrepancy occur?
It is important to note that the errors reported in Table 2 represent only the statistical uncertainty derived from the noise of our data.

The overall error is likely dominated by systematic uncertainties, which weren't included in our calculations, such as the uncertainty on the galaxy's distance and the intrinsic scatter of the empirical M/L relations. Including these systematic uncertanties would broaden the error bars significantly, likely reconciling the discrepancies observed between the different band methods.
For example, according to Bell et al. (2003), the correlations between color and M/L ratio are characterized by an intrinsic scatter of ≈ 0.1 dex, which corresponds to a systematic error of approximately 25% (the gray line in figure 5). Just by considering this systematic error, we notice that not only the result for the two bands now overlap, but also the results for the different color indeces.

4 DISCREPANCY BETWEEN (G-I) AND (G-R)

Systematic errors that we haven't considered might make the two mass estimates compatible. However, since the two estimates rely on essentially different spectral regions, given that $(g-i)$ covers a larger wavelength range than $(g-r)$, it is natural to apply a deeper analysis on the discrepancy between the two results. The fact that we obtain two distinct mass values is unlikely to be attributable to systematic errors only. Instead, the $(g-i)$ and $(g-r)$ colors presents different properties that cause them to respond differently, ultimately affecting the mass estimation.
In this section, we will focus on what makes the use of the r filter different from the use of the i filter and we will analyze the trade-off between the two methods to understand which result is physically more reliable.

4.1 Quantum Efficiency

The r filter gives a medium mass of around $5.88 \cdot 10^{10} M_{\odot}$. What are the positive aspects of choosing to compute the mass through this band?
The main characteristic that makes this measure reliable is that the r band presents a high instrumental efficiency. Looking at the transmission curves of both the r filter and the i filter (figure 1), it's immediately clear how the quantum efficiency of the r band is higher compared to the i band: consequently, for a given exposure time, the r band collects more electrons per incident photon. This generally guaranties a high S/N, resulting in cleaner photometry with lower statistical errors. Moreover, the night sky in the infrared region is significantly more luminous and variable, due to atmospheric emission lines (fringing), creating a background noise that is harder to subtract. This was evident during the data reduction and sky subtraction process. Data coming from the r filter were easier to handle, while data from the i filter were very noisy and required a more careful sky subtraction: as mentioned in 2.3, we had to crop the frames of the i filter before applying the sky subtraction because the background on the edges was too anomalous.

4.2 Dust Absorption

However, looking at the results in Table 2, the error associated to the $(g-i)$ measure of the mass is surprisingly smaller than the $(g-r)$ one, despite the weakest quantum efficiency we just mentioned. This, at first glance, is a counter-intuitive observation that can however be explained taking into account the physical properties of the galaxy, specifically the presence of dust. Since NGC 6946 is a high Star Formation Rate galaxy, it contains a non negligible dust component that attenuates the stellar flux.

According to Calzetti (2001)'s law, dust absorbs light selectively depending on the wavelength. For the r and i bands, the dust absorption coefficients are respectively

- $A_r = 3.52 \cdot$ E(B-V)
- $A_i = 2.75 \cdot$ E(B-V)

where E(B-V) represents the color excess due to dust. As we can see, the r band suffers more dust attenuation compared to the i band. Part of the stellar luminosity is absorbed and doesn't reach the telescope. This might lead to an underestimate of the luminosity and consequently of the stellar mass.
On the other hand, the i-band is more transparent to dust. This allows us to detect the more stellar mass, explaining why the $(g-i)$ estimate is higher.

5 CONCLUSIONS

In this work, we presented a complete photometric analysis of the galaxy NGC 6946, using data from the TOBi telescope. Starting from raw images we learnt how to reduce, calibrate and analyze them to derive a fundamental property of the galaxy: its total stellar mass.
We computed the mass using two different colors, finding a discrepancy between the results: $M \approx 5.88 \cdot 10^{10} M_{\odot}$ using $(g-r)$ and $M \approx 6.81 \cdot 10^{10} M_{\odot}$ using $(g-i)$. Our discussion highlighted an important trade-off between instrumental precision and physical accuracy. From the instrumental perspective, the measure derived from the $(g-r)$ color is technically superior. As discussed in the previous section, the r band benefits from a higher Quantum Efficiency and a lower sky background noise compared to the i band. This results in a measurement with high S/N and statistical precision. However, from a physical perspective, the r band flux is absorbed by dust more than the i band is, leading to an underestimation of the true stellar mass.
In conclusion, if the goal is to minimize statistical errors and work with the highest quality data available from the instrument, the $(g-r)$ estimator is preferable. However, we must be aware that the value we obtain can be an underestimate compared to the real stellar mass.

REFERENCES

Bell E. F., McIntosh D. H., Katz N., Weinberg M. D., 2003, The Astrophysical Journal, 585, L117
Calzetti D., 2001, Publications of the Astronomical Society of the Pacific, 113, 1449
van Leeuwen F., et al., 2022, Gaia DR3 documentation